

Flood Polynomial Model (MKM-FPO-001)

Wolfram Oracle Benchmark of v2.0 and its GBM Approximation

MKM Research Labs

Version 0.2 — 03-November-2026

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1 Executive Summary

This document benchmarks the Flood Polynomial Model (MKM-FPO-001), v2.0, a closed-form surrogate for the per-gauge GBM flood classifiers (MKM-FC-001) that the stress scenarios use to price P(flood) hour by hour. It asks whether the code computes what its model document (`flood_poly_v2.0.pdf`) says, whether the document's claims hold, and how well the model approximates the classifiers.

Version 0.1 of this document benchmarked v1.0 and found it fell as the water rose, and that none of its document's tables or accuracy claims held. v2.0 is a new model, refitted to every gauge, and this is its benchmark.

The code computes its documented model. `p_flood_poly` agrees with a 50-digit implementation of the document's equations, printed coefficients and printed envelope, through the floor, the held hours and the step at severe; so do `p_flood_series`, the sensitivity module's evaluation and its slope. The port stress screen's JavaScript, run under node on the model the server sends it, agrees too (Table 1). Wolfram, from the exact coefficients and knots, agrees with the reference to 3.1×10^{-31}.

The document's claims hold (11 of 11, Table 3).

- **Its tables come from the code.** The code reproduces 88 of the 88 cells of the probability table, and marks the same 23 cells as held. The coefficients and envelope are the code's.
- **The envelope is the data's.** Recomputed from the storm catalogue, every printed knot is within 0.05 hours of where the storms first reach that level.
- **Rising water never lowers P.** The code over every hour finds no fall (largest 0.0e+00). Between the fit's grid levels, Wolfram's smallest rate of rise is 0.0057 at any whole hour where it runs free, and 0.0105 along the envelope where it is held. The model's hours are whole; between hour 0 and hour 1 the kernel's slope does dip below zero, as the document now says, but no hour falls there.
- **The accuracy table reproduces.** Recomputed with the document's model on the fit's own held-out storms, every row matches.

The approximation, measured. On every real storm state of the 11 gauges (Table 2):

- below severe, the model is within 0.0066 to 0.0217 of the GBM on average;
- near severe (75–100%), the mean error is 0.043 to 0.069, and 11 to 22% of states are off by more than 0.10;
- the bias near severe ranges from -0.033 to +0.028.

That is the document's own account: close on average, imprecise state by state, because the GBMs see the rate of rise and the model does not.

Evidence. 10 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `5ef357e130c7`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-11-03 by `python -m tools.wolfram_oracle fpo001`.

2 Model Purpose and Scope

2.1 What is benchmarked

- `src/models/stress/flood_poly.py`: `p_flood_poly`, `p_flood_series`, `p_flood_simple` and the model the server sends the browser.
- The sensitivity module's evaluation and slope, in `src/models/stress/sensitivity.py`.
- The port stress screen's evaluation, in `src/static/js/trading/port_stress/setup.js`.
- The model against the production GBM classifier of every gauge that has one.

2.2 What is not

- The classifiers themselves (MKM-FC-001).
- The stress scenarios' hydrographs, and the P&L built on P.
- Whether the fit is the best a cubic can do: the benchmark tests the fitted model as documented, not the optimiser.

This is a benchmark of the implementation and of the approximation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

The document's model is

$$P = \begin{cases} 1 & w \geq s, \\ \sigma\big(z(h, \max(t, \tau(h)))\big) & w < s, \end{cases} \qquad z = \sum_{k=1}^{10} c_k h^{i_k} t^{j_k}, \quad h = \ln \max(w/s, 0.2), \quad t = \ln \frac{\text{hour} + 1}{168}, \tag{1}$$

with the exponent clamped to ± 30, and τ the support envelope through the document's printed knots, linear in t between them. The reference evaluates it at 50 digits from the printed coefficients and knots.

What the polynomial can know. Each gauge's GBM also sees the rate of rise Δh and its change. A function of (h, t) alone can at best reproduce the GBM's average over the storm states that share a level and an hour. The approximation is therefore judged on those real states, as production meets them.

4 Benchmark Method

The code. The code is checked against the reference over three severe thresholds, 21 level ratios from zero to 1.5 (through the floor, the envelope's knots and severe), and 19 hours (through the envelope). The screen's JavaScript is run under node on the same cases, with the model the server sends it.

Wolfram. From exact coefficients and knots, Wolfram evaluates P at every cell of the document's probability table below severe, with the hold written as its own `Piecewise`; $\partial P/\partial h$ at a fixed hour symbolically, at held and free points; and, by `NMinimize`, the smallest $\partial z/\partial h$ at each of the 168 whole

hours over the levels that hour is not held at (Wolfram inverts the envelope itself), and the smallest rate along each segment of the envelope. The fit holds those rates positive only at grid points; this checks between them. Each minimum is checked against the reference at Wolfram's point, and against a dense grid, so a local minimum would not pass.

The GBM comparison.

- For every gauge with a classifier, 2,000 storms are drawn once (seed 7) from the catalogue, and each storm's states are rebuilt through production's training-vector builder. The model and the GBM are compared on the states below severe.
- The same states give the highest level any storm reached by each hour, and from it the hour each printed envelope level is first reached.
- The document's held-out table is recomputed on the fit's own split, with the document's model rather than the code.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Eq. 3: `p_flood_poly` against the reference, worst relative error	3.8×10^{-14}	1.0×10^{-12}	pass
A threshold at or below zero gives zero	yes		pass
`p_flood_series` is `p_flood_poly` hour by hour	0	0	pass
`p_flood_simple` is $\min(1, w/s)$	7.3×10^{-17}	1.0×10^{-12}	pass
The model as `stress.sensitivity` evaluates it	3.8×10^{-14}	1.0×10^{-12}	pass
Its closed-form $\partial P/\partial h$, held hours included	5.1×10^{-13}	1.0×10^{-12}	pass
The server sends the screen the document's coefficients and envelope	yes		pass
The screen carries no coefficients of its own	yes		pass
The screen's JavaScript, run under node, against the reference	6.8×10^{-14}	1.0×10^{-12}	pass
Wolfram agrees with the reference (table cells, slopes, minima at their points)	3.1×10^{-31}	1.0×10^{-25}	pass

5.1 The gauges

Table 2: The model against each gauge's GBM on every real storm state, near severe ($w/s \geq 0.75$) and below it

Gauge	States	GBM	Poly	Mean $\lvert d \rvert$	95%	> 0.1	All below
GAUGE-b2dbe58b	14,592	0.326	0.349	0.060	0.209	21%	0.0118
GAUGE-dc5e88fb	9,894	0.296	0.288	0.043	0.153	11%	0.0085
GAUGE-1aaf8b08	18,815	0.413	0.427	0.057	0.192	17%	0.0204
GAUGE-d208bf3d	14,990	0.445	0.412	0.069	0.246	22%	0.0217
GAUGE-c0fbc443	10,256	0.259	0.278	0.048	0.172	15%	0.0074
GAUGE-a6be4ee9	11,406	0.300	0.309	0.050	0.174	16%	0.0099
GAUGE-ef53398c	8,379	0.281	0.277	0.048	0.187	14%	0.0066
GAUGE-085522ee	16,268	0.416	0.409	0.057	0.194	17%	0.0176
GAUGE-9fffb377	12,730	0.287	0.316	0.057	0.201	20%	0.0100
GAUGE-f8629444	14,105	0.406	0.379	0.060	0.220	19%	0.0173
GAUGE-9e5f3ef9	17,404	0.374	0.394	0.058	0.194	19%	0.0167

5.2 The model document's claims

Table 3: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
§4.3, Table 2	the ten terms and coefficients	the code's, to the printed six decimals	met
§3.3, Table 1	the earliest hour any storm reached each level	the code's knots; recomputed from the catalogue, every knot within 0.05 h (limit 0.1); first level at hour 0 0.326	met
§7.1, Table 4	P by level and hour, held cells marked	88 of 88 cells reproduced by the code; the 23 held cells marked as the code holds them	met
§3.2	$P = 1$ at or above severe; constant below the floor	so at every hour	met
§3.2	below the floor, effectively zero	at most 0.0012	met
§3.4	rising water never lowers P (the code)	over 2,000 levels at each of 168 hours, the largest fall is 0.0e+00	met
§3.4	rising water never lowers P (Wolfram, between the fit's grid levels)	smallest $\partial z/\partial h$ at any whole hour where it runs free 0.0057; along the envelope 0.0105	met
§3.2	from hour 16 on P never rises with the hour; before it, by at most 0.06	largest rise from hour 16 0.0e+00; before it 0.055	met
§3.1	$t \approx -5.12$ at hour 0	-5.1240	met
§5, Table 3	accuracy on the held-out storms, by gauge	11 of 11 rows reproduced with the document's model, on 1,807,233 fitted and 1,807,896 held-out states	met
§2.3	the scenario holds $P = 1$ once severe is reached	so the route does	met

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The GBMs are the reference here; whether they are right is MKM-FC-001's question.

2. **The accuracy is measured, not improved.** The document reports the error near severe and its cause; the benchmark confirms the figures, which remain large state by state.

3. **The envelope is the catalogue's.** It is reproduced from the same storms it was measured on. A storm faster than any in the catalogue is priced at the envelope, as the document says.

7 Document History

Date	Version	Description	Author
03-Nov-2026	0.1	First issue: tests v1.0	David K Kelly
03-Nov-2026	0.2	Tests v2.0, the refitted model	David K Kelly